An Interactive Generative AI Tool to Help Teachers Contextually Customize Scratch Projects

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Abstract

Teachers are well positioned to customize curricula to local contexts. In CS, one approach involves choosing themes relevant to students and incorporating them into the technical materials (e.g., Scratch projects, CS concept explanations). However, teachers' time constraints make curriculum customization challenging. This poster introduces Conjuror, an interactive GenAI tool to help teachers create contextually customized Scratch projects aligned with a structured curriculum while retaining teacher agency. We present Conjuror design and its pilot with two teacher cohorts, showing promising evidence for process efficiency and output quality.

CCS Concepts

• **Social and professional topics** → **K-12 education**; • **Human-centered computing** → **Interactive systems and tools**.

Keywords

curriculum customization, AI tools, Scratch programming

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1 Introduction and Related Work

Many recent works have explored using GenAI to *help **college CS instructors** create course materials*, both in bulk [4, 6] and on-demand [3, 5]. However, much less work has studied using GenAI to *help **K-12 CS teachers** in course preparation.* GPT has been used to adapt Scratch projects to new contexts, producing creative but culturally insensitive or overly complex ideas [7]. It has also been tested for generating solutions and auto-grading Scratch projects, but often fails syntaxes and produces inaccurate scores [2].

This poster presents the design and pilot of Conjuror, an interactive tool that leverages GPT and Stable Diffusion to help teachers create contextually customized Use-Modify Scratch projects[1] that align with an intermediate CS curriculum [1]. It provides structure and support throughout the process (from brainstorming a project idea to creating a fully functional Scratch project and associated student-facing worksheets). Our goals are to support teachers in:

(1) Producing high-quality materials in a short time (< 30 mins) so it is feasible within normal class preparation time, while
(2) Leveraging teachers' localized expertise about their students, schools, and local communities, and
(3) Allowing teachers to remain in control of how and where AI-generated content is adopted.

2 Methods

2.1 Tool Design

Conjuror has four main components, each guides teachers through inputs, decisions, and ideations (Figure 1). Teachers may select, modify, or replace Conjuror suggestions with their own. All AI prompts were pre-built by researchers and tailored to Scratch Encore.

ThemeStormer suggests 10 themes related to students' family, local community, school, and youth culture. For example, for local community culture, it asks the teacher the city and state of their school, then generates themes about local events and landmarks. For family culture, it uses inputs about students' ethnicities and originating countries to suggest cultural holidays and food themes.

ProjectStormer suggests four project ideas that satisfy the technical attributes of base projects from a Scratch Encore module and relate to the chosen theme. Each idea includes a title and descriptions of the backdrop and each sprite's attributes and actions.

PicStormer first suggests sprites and backdrops from the Scratch Library. The teacher can either choose these or generate bespoke images. Generated sprites can be humans, objects, or animals. The teacher can select various modifiers for gender, age group, and skin/hair/eye color, to best fit the demographic of their students.

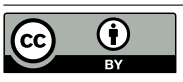
ProjectCreator swaps out the sprites and backdrop in the Scratch Encore base project and outputs a Scratch project file. It reuses the base project's code; only the images are changed. The teacher can upload the project file to Scratch and adjust image sizes and sprites' starting location. The tool also updates the associated student worksheets (in Google Docs and Forms) and email them to the teacher.

2.2 Tool Pilot

We piloted Conjuror with 15 CS teachers (teaching students ages 9-13) in the U.S and collected data from: (1) tool usage logs, (2) teacher surveys, and (3) semi-structured teacher interviews.

Six teachers participated in Pilot #1, conducted on the final day of a week-long online summer professional development aimed at customizing Scratch Encore. Participants observed a facilitator walk through using Conjuror, then attempted to use it by themselves.

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[1]Sample projects given to students to explore and modify the code.



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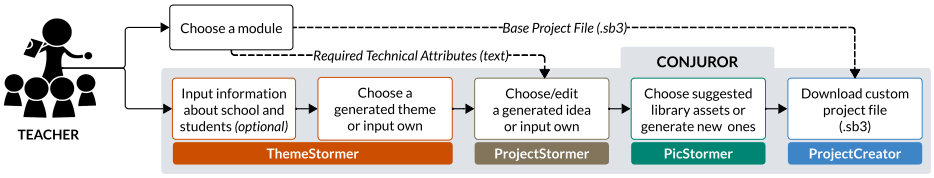


Figure 1: Conjuror project customization pipeline

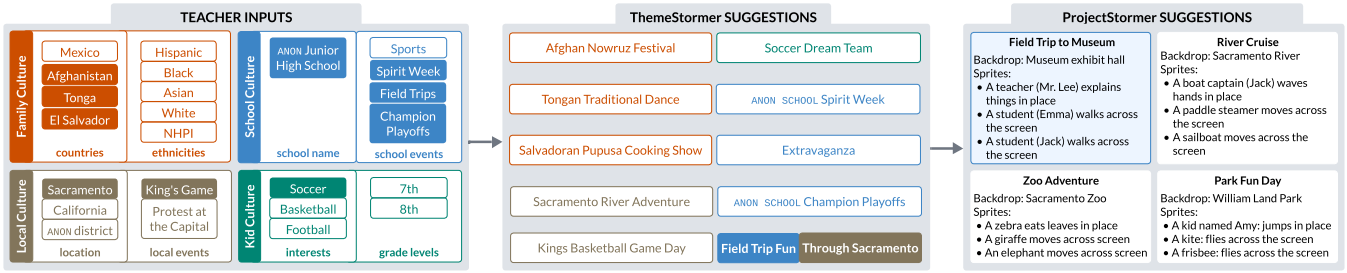


Figure 2: Charting the flow of information through a pilot teacher's interaction with ThemeStormer and ProjectStormer.

Nine teachers who had trained in Scratch Encore from various providers and indicated interest in trying out Conjuror participated in Pilot #2, taking place during a school year. They were given a tutorial video and instructed to use the tool at their leisure.

3 Results Overview

3.1 Tool Usage

Our logged data showed that all teachers finished customizing in under 20 minutes (avg. 10.02). Most time was spent in PicStormer (avg. 5.66), where teachers selected and/or generated multiple images for a backdrop and multiple sprites. Time spent on ThemeStormer and ProjectStormer averaged 2.4 and 1.95 minutes respectively.

Looking at how often teachers chose from different sources, we found that teachers chose Conjuror-generated themes slightly more often than their own (62%) while almost always choosing Conjuror-generated project ideas (93%). For sprites and backdrops, they balanced Scratch Library (43%) and AI-generated images (57%).

3.2 Teacher Feedback

Teachers found Conjuror to be a time-saver: *"The project idea and image generation saves so much time. As teachers, we don't have a lot of time so that was so key to me."* One said they could quickly create multiple projects for different demographics and let students pick one that is most relevant to them. Teachers also valued how Conjuror could expand their inputs to creative ideas: *"I put in kid culture and the city I was in, and it spit out a gym shoes parade in Chicago street... I would never thought about that."* Finally, teachers appreciated that the tool allowed them to customize human sprites: *"You could get down to the people's skin tone and eye color... I like that all of the characters were brown or people of color."*.

Besides positive feedback, teachers found Conjuror somewhat difficult to navigate at first and requested more step-by-step guidance. Some noted mismatches between requested and generated images (e.g., a teacher wanted *"a soccer goal without a ball"* but got images with one). While PicStormer has an optional field for specifying image details, not all teachers noticed it and one who did suggested providing "*supplemental training on prompt tips*".

4 Contribution and Future Work

With Conjuror, we take a responsible first step in bringing GenAI into K-12 classrooms, positioning it as a productivity tool for teachers rather than a student-facing solution. It both builds on teachers' knowledge about their local communities and allows them to remain in control of how and where AI-generated content is adopted.

In future work, we will provide a comprehensive user guide and training, followed by a classroom study to examine teacher experiences using a streamlined GenAI tool for CS class preparation and most importantly, its impacts on student engagement and learning.

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